ELSEVIER

Tectonophysics 326 (2000) 57–72

TECTONOPHYSICS

www.elsevier.com/locate/tecto

Paleomagnetism and geobarometry of the Big Creek Batholith suggests that the Yukon–Tanana Terrane has been a parautochthon since Early Jurassic

D.T.A. Symons [a,*], P.R. Williams [a], P.J.A. McCausland [b], M.J. Harris [a], C.J.R. Hart [c], W.H. Blackburn [d]

[a] *Earth Sciences, University of Windsor, Windsor, ON, N9B 3P4, Canada*
[b] *Department of Earth Sciences, University of Western Ontario, London ON, N6A 5B7, Canada*
[c] *Yukon Geology Program, Government of Yukon, Box 2703 [F-3], Whitehorse, YT, Y1A 2C6, Canada*
[d] *Royal Roads University, Victoria, BC, V9B 5Y2, Canada*

Abstract

The $\sim 190 \pm 5$ Ma Big Creek hornblende-syenite and monzogabbro batholith has an outcrop area of ~ 400 km^2 within the Yukon–Tanana Terrane (YTT) in the west-central part of the Yukon Territory of Canada. Paleomagnetic analysis of 245 specimens from 21 sites isolates a characteristic remanent magnetization residing principally in magnetite and subordinately in hematite. The unit mean direction for the batholith from 16 normal polarity and two reverse polarity sites is $D = 305.6^\circ$, $I = 72.1^\circ$ ($\alpha_{95} = 3.9^\circ$). Al-in-hornblende geothermobarometry at six locations yields emplacement depths of $\sim 16 \pm 2$ km. Three arguments suggest that the characteristic remanent magnetization was acquired on exhumation at ~ 180 Ma. More significantly, the pole position indicates no northward translation but counterclockwise rotation, implying that the YTT formed close to its present position on the Jurassic margin of the North American craton. This implies, in turn, that the allochthonous terranes of the Intermontane and Coast belts of the western Cordillera impacted onto, and overrode, the YTT upon accretion during Late Cretaceous to Middle Eocene time. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Big Creek Batholith; geobarometry; paleomagnetism; Yukon–Tanana Terrane; Canada; Jurrasic; tectonics

1. Introduction

The Yukon–Tanana Terrane (YTT) underlies a large part of eastern Alaska, south-central Yukon and northernmost British Columbia (Fig. 1). Much of the YTT in eastern Alaska and west-central Yukon is unglaciated, resulting in limited exposures of poor quality, which has hampered mapping. The YTT has had a complex geological evolution that is recorded in multiple metamorphic and deformational events (Mortensen, 1992), and it contains an assemblage of Paleozoic rocks with apparent North American stratigraphic affinity (Tempelman-Kluit, 1976) that could be: (1) a parautochthon of North America against which the amalgamated Intermontane terranes impacted and accreted (McClelland et al., 1992; Mihalynuk et al., 1994); (2) an allochthonous terrane that has travelled some 1500 km northward along the margin of the North American craton together

* Corresponding author. Tel.: +1-519-253-3000; fax: +1-519-973-7081.
E-mail address: dsymons@uwindsor.ca (D.T.A. Symons)

0040-1951/00/$ - see front matter © 2000 Elsevier Science B.V. All rights reserved.
PII: S0040-1951(00)00146-3

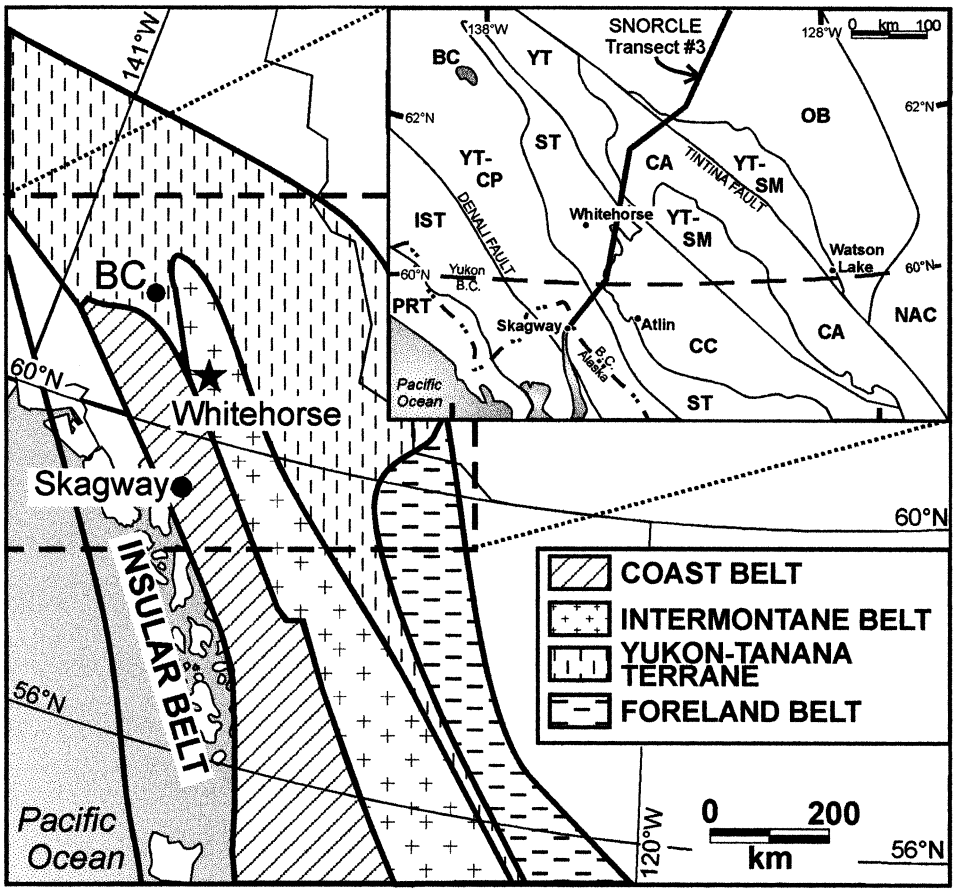


Fig. 1. Morphogeological belts and tectonic terranes (inset) of the northern Canadian Cordillera. Abbreviations are: BC: Big Creek batholith; PRT: Pacific Rim terranes; IST: Insular terrane; YT-CP: Yukon–Tanana–Coast Plutonic terrane; ST: Stikine terrane; YT: Yukon: Tanana terrane; YT-SM: Yukon–Tanana: Slide Mountain terrane; CA: Cassiar terrane; OB: Omineca Belt; NAC: North American craton.

with the amalgamated Intermontane Belt terranes (Irving et al., 1996; Johnston et al., 1996b; Harris et al., 1997, 1999); (3) an allochthonous terrane that originated somewhere offshore in the Pacific basin and was translated and accreted onto North America (Hansen, 1990; Hansen and Dusel-Bacon, 1998); or (4) an assembly of microterranes that contains elements of (1), (2) and/or (3) (Hansen et al., 1991; Mortensen, 1992). Further, the YTT has been ignored in most geotectonic models, excluding Mihalynuk et al. (1994), because of these uncertainties, although Wynne et al. (1998) accept that the YTT and Stikine terranes had a shared history since the Early Jurassic. This paper presents paleomagnetic data from the Early Jurassic Big Creek batholith in the central part of the YTT that strongly favors a parautochthon model for the YTT's place in the geotectonic accretionary assembly of the northern Cordillera since Early Jurassic.

2. Geologic setting

The Yukon–Tanana Terrane is composed of three lithological packages: a pre-Devonian passive margin assemblage of pelitic to quartzose metasedimentary rocks with minor marble and rare

amphibolite (Nisling assemblage), a Devono-Mississippian continental arc assemblage of mafic and felsic volcanics with associated orthogneisses (Pelly gneiss) and graphitic siliciclastics (Nasina assemblage), and a less-extensive Permian felsic volcanic package (Klondike schist) with associated orthogneisses (Mortensen and Jilson, 1985). Polyphase deformation and associated metamorphism has largely obscured depositional features and contact relations between the units. Late Triassic to Early Jurassic batholiths including the Big Creek batholith are undeformed and intruded the YTT after its last regional deformation event. Post-Early Jurassic uplift and extension has been suggested for many parts of the Yukon–Tanana Terrane (Hansen, 1990; Hansen et al., 1991; Pavlis et al., 1993; Hansen and Dusel-Bacon, 1998) but cannot be documented for the study area where rocks as old as Triassic remain undeformed and most reset isotopic systems record Jurassic cooling.

The ~190 Ma Big Creek batholith is located about 60 km west of Carmacks or about 200 km northwest of Whitehorse in the Yukon (Fig. 1). It is exposed over an area of ~400 km^2, has a northwest–southeast (NW–SE) length of ~55 km (Fig. 2) and was emplaced into older rocks of the YTT basement (Tempelman-Kluit, 1984). Locally, the Big Creek batholith was subsequently intruded by the ~105 Ma Early Cretaceous Dawson Range batholith and overlain by the ~100 Ma Late Cretaceous Mount Nansen Group comprising predominantly andesitic volcanic rocks (Tempelman-Kluit et al., 1991). Thereafter, the batholith was covered by basaltic flows and tuffs of the ~70 Ma Late Cretaceous Carmacks Group. At ~69 Ma, the Seymour Creek stock was emplaced into the batholith and was subsequently cut by the major NW–SE-trending Big Creek Fault.

The Big Creek batholith is composed of medium- to very coarse-grained hornblende-syenite and monzogabbro with megacrystic alkali feldspar and hornblende crystals. Petrologically variable, it ranges from cumulate zones of hornblendite and gabbro to true syenite and granite (Carlson, 1987; McInnes, 1987; Payne et al., 1987). It is generally massive but does have some magmatically and tectonically foliated zones.

Several lines of geologic evidence suggest that the batholith has not been tilted significantly since emplacement: (1) regional metamorphic grades in the Big Creek's host rocks are constant around the pluton; (2) west to northwest trending joint sets dip vertically on average; (3) younger felsic dikes dip vertically on average within the batholith; (4) overlying ~70 Ma Carmacks volcanics at the batholith's north and south ends are roughly horizontal on average; (5) primary foliation orientations in the north and potassium feldspar-hornblende cumulates in the south both indicate levels near the base of the batholith; and (6) contact foliations dip roughly vertically on average (C.J.R. Hart, pers. commun.).

An average emplacement age of 190 ± 5 Ma has been defined by four isotopic dates for the Big Creek batholith, including U–Pb ages from zircon of 197 ± 1 Ma and from titanite of 184 ± 7 Ma (J. Mortensen, pers. commun., 1999), a K–Ar hornblende age of 188.8 ± 3.8 Ma (Hunt and Roddick, 1987) and a K–Ar hornblende age of 184 ± 7 Ma (Stevens et al., 1982). Younger K–Ar hornblende ages of 152 ± 7 and 142 ± 10 Ma (Stevens et al., 1982) and a Rb–Sr whole-rock isochron age of 138 ± 10 Ma (McInnes, 1987) indicate variable thermal resetting proximal to Cretaceous intrusions.

3. Paleomagnetic methods

Twenty-one sites were sampled by helicopter from the bases of tors of the Big Creek batholith to represent it both spatially and petrographically (Fig. 2). At least six cores were drilled at each site and oriented in situ using a solar compass. Two specimens were sliced from most cores, yielding a total collection of 245 specimens. They were stored in a magnetically shielded room in which all paleomagnetic measurements were done using an automated Canadian Thin Films CTF DRM-420 cryogenic magnetometer, a Sapphire Instruments SI-4 alternating field (AF) demagnetizer, a Magnetic Measurements MMTD-1 thermal demagnetizer, and a Sapphire Instruments SI-6 pulse magnetizer.

Following measurement of the natural remanence magnetization (NRM) of the specimens,

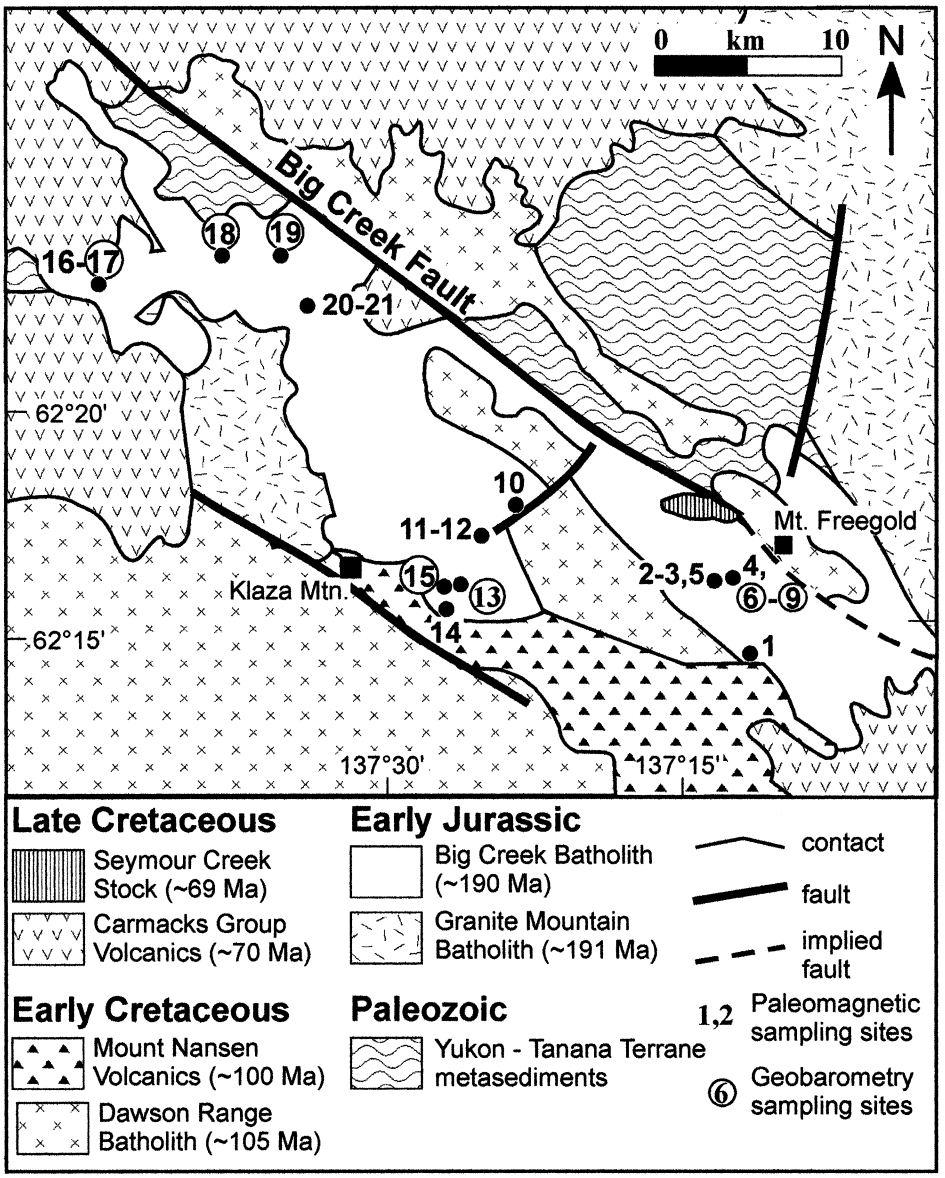


Fig. 2. Geology of the Big Creek batholith (after Tempelman-Kluit, 1984; Carlson, 1987; Payne et al., 1987) with sampling sites.

two pilot specimens per site were AF-demagnetized in 10 steps up to 130 mT, and two more were thermally demagnetized in 16 steps up to 690°C. The thermal steps were chosen in part to check for unblocking in the diagnostic 80–130, 260–325, 520–585, and 620–675°C temperature ranges that are indicative of goethite, pyrrhotite, magnetite and hematite, respectively. Based on the pilot specimen results for each site, the remaining specimens were AF-demagnetized in five or six steps spanning the range from 20 to 85 ± 25 mT. This was followed by three or more additional thermal steps up to 655°C or AF steps up to 130 mT. The characteristic remanent magnetization (ChRM) vector for each specimen was obtained using the Kirschvink (1980) least-squares method over three

or more consecutive steps, requiring that the maximum angular deviation not exceed 15°, although it was usually <5°. Site and unit mean ChRM directions were obtained following Fisher (1953). Saturation isothermal remanent magnetization (SIRM) acquisition curves were obtained for 12 specimens by pulse-magnetizing them in 11 steps up to 900 mT, and the SIRM demagnetizing curves were obtained by AF-demagnetizing the specimens in seven steps up to 140 mT.

3.1. Paleomagnetic results

As expected for coarse-grained igneous rocks that are rich in mafic minerals, the specimens from the Big Creek batholith have relatively intense NRMs that mostly exceed 1×10^{-2} A/m.

The intensity decay curves on thermal step demagnetization (Fig. 3) show that goethite is not a significant remanence carrier, despite the problem of deeply oxidized rock in this unglaciated area. The curves also show that pyrrhotite is not a significant carrier. Rather, they show that magnetite is the main carrier having distributed unblocking up to ~500°C, followed by a substantial unblocking between ~500 and ~585°C. In a few cases, there is additional unblocking between 600 and 675°C that indicates the presence of some hematite. The increased intensities in some specimens above 600°C are attributed to the creation of new magnetic mineral phases from chemical reactions in the oven. Thermal demagnetization typically removes the viscous remanent magnetization (VRM) by about 270°C and gives well-defined ChRM directions only in a minority of specimens (Fig. 4e and f). AF step demagnetization typically removes the VRM rapidly by 10 or 20 mT, with the ChRM being well isolated in the last ~20% of the NRM (Fig. 4a–d). The SIRM curves show clearly that the ChRM is carried by pseudosingle to multidomain magnetite (Fig. 5), consistent with the thermal demagnetization results.

Isolation of well-defined ChRM endpoint directions was possible for two-thirds of the specimens using the least-squares method (Kirschvink, 1980). However, for the remaining one-third — mostly those with a very large VRM component relative to their ChRM component — it was necessary to

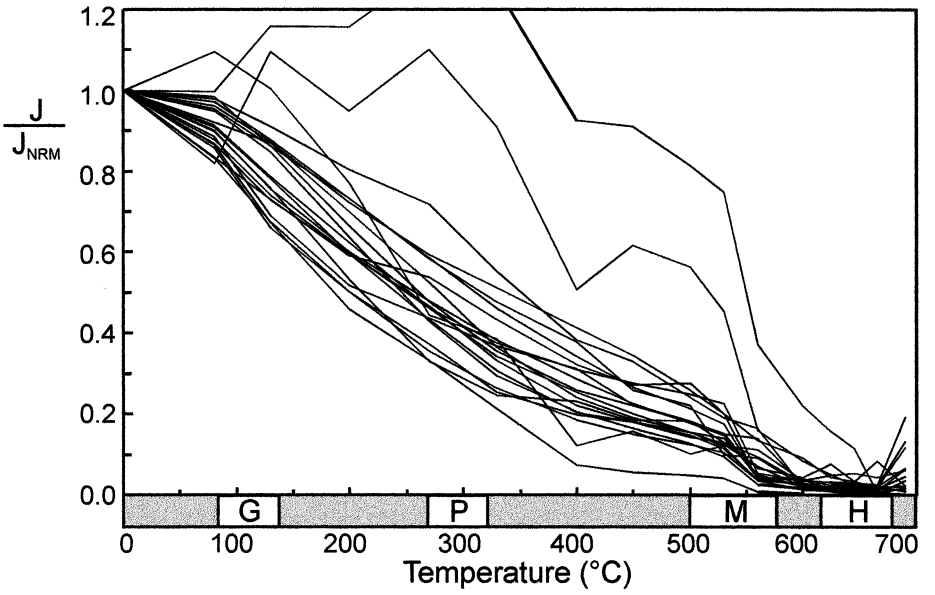


Fig. 3. Thermal demagnetization intensity decay curves for the Big Creek batholith. The preferred unblocking temperature ranges diagnostic of goethite, pyrrhotite, magnetite and hematite are indicated at the bottom by the G, P, M and H, respectively. The vertical axis is the ratio of the measured intensity to the NRM intensity.

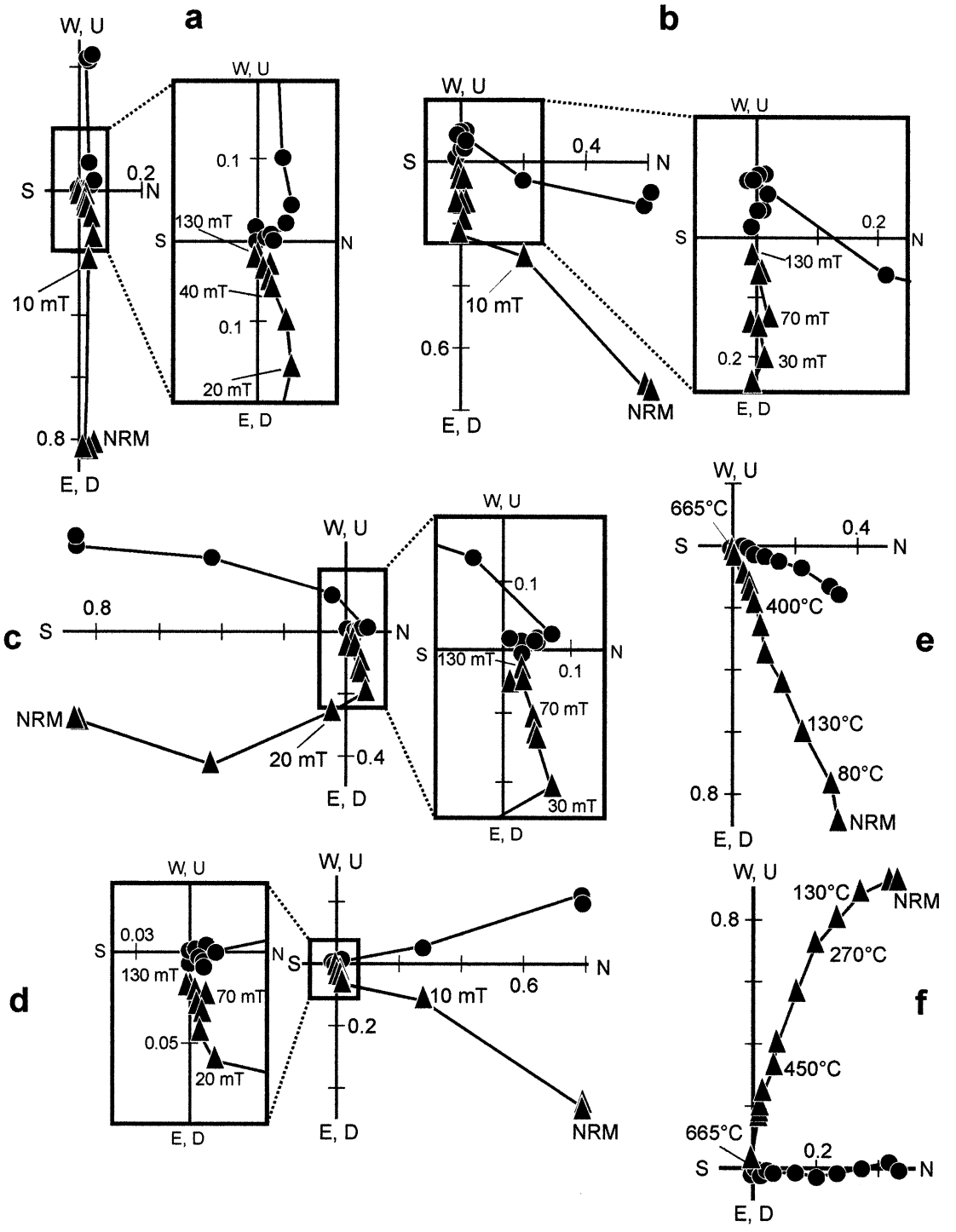


Fig. 4. Representative orthogonal step demagnetization plots (Zijderveld, 1967) for example specimens from sites: (a) #1, (b) #8, (c) #12, (d) #18, (e) #7 and, (f) #17. The circles are in the north–east–south–west (N–E–S–W) plane, and the triangles are in the up–north–down–south (U–N–D–S) plane. Axial values give the ratio of the measured to NRM intensity.

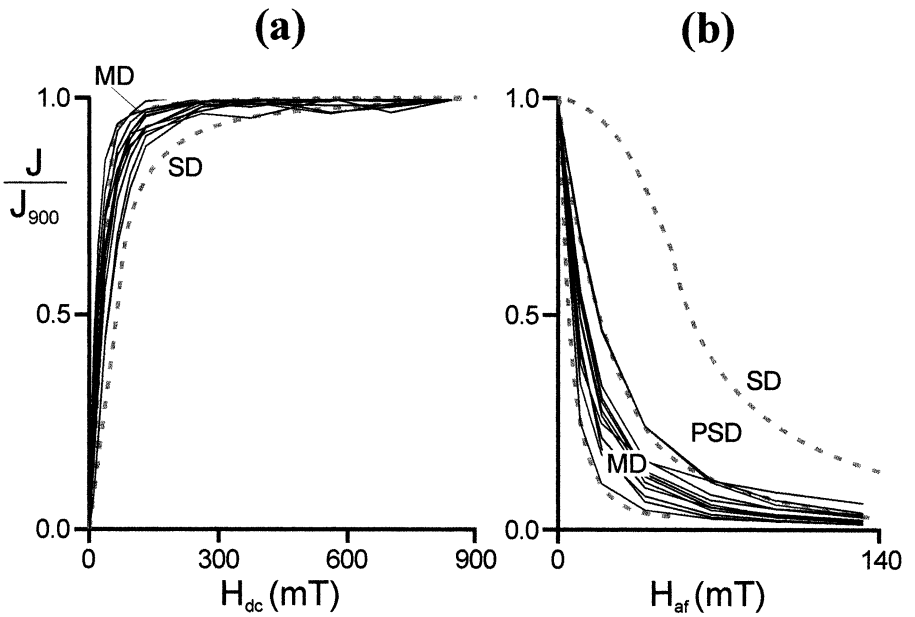


Fig. 5. SIRM acquisition and decay curves for the Big Creek batholith. The H_{dc} and H_{af} axes are measured in milliTesla (mT). The vertical axis is the ratio of the measured (J) to saturation (J_{900}) intensity at 900 mT. The dashed type curves are for multidomain (MD), pseudosingle domain (PSD) and single-domain (SD) magnetite.

use the remagnetization circle method of Halls (1978) to obtain an 'endpoint'. Site mean ChRM directions were determined for 18 of the 21 sites, including 16 with normal polarity and two (sites 3 and 17) with reversed polarity (Table 1). For each site, the ChRM was found to be present over a broad range of AF coercivities but, when identifiable, generally only in the high unblocking temperature range (Table 1). The latter suggests that the hematite-borne ChRM with a direction divergent from the Earth's present magnetic field, is not a modern weathering component.

When the site mean ChRM directions are plotted on a stereogram (Fig. 6), it is apparent that the two reversely polarized sites are antiparallel to the normally polarized population. Thus, the former were reported to their antipodal normal position and a unit mean ChRM direction of declination (D)=305.6°, inclination (I)=72.1°, radius of cone of 95% confidence (α_{95})=3.9°, and precision parameter (k)=82, was calculated for the Big Creek batholith. This ChRM direction yields a paleopole at 62.9°N, 146.6°E (d_p=6.1°, d_m=6.9°).

4. Geobarometric methods and results

The aluminum-in-hornblende geobarometer and amphibole-plagioclase geothermometer were used to estimate the pressure and temperature of crystallization in order to establish paleohorizontal control and post-crystallization tilt. The theory and procedures have been described in detail elsewhere (Blundy and Holland, 1990; Anderson and Smith, 1995; Harris, 1998; Harris et al., 1999). Following petrographic examination of a thin section from each site, specimens from seven sites were chosen to represent the batholith spatially. Chemical analyses were made for hornblende and plagioclase pairs on a JEOL JXA-8600 Superprobe at the University of Western Ontario using a probe current of 10 nA, accelerating voltage of 15 kV, beam focus of 2 μm for amphibole and 5 μm for plagioclase and a 20 s accumulation time. The calibration standards for amphibole were: Si — Cr-augite, USNM 164905; Al — anorthite; Cr — chromite, USNM 117075; Fe — fayalite; Mn — rhodonite; Mg — olivine, USNM 111312; Ca — Wakefield diopside; Na — Amelia albite; K–ortho-

Table 1
Site and unit means of paleomagnetic data for the Big Creek batholith[a]

Site	Sampling				Demagnetization steps		Mean remanence direction			
	N	*n*	*e*	*r*	AF (mT)	Thermal (°C)	*D* (°)	*I* (°)	α_{95} (°)	*k*
01	11	11	11	0	20–130	200–665	329.1	61.6	7.4	39
02	8	5	3	2	40–100	–	279.3	77.9	11.3	47
03[b]	9	9	4	5	45–80	560–665	291.5	79.5	6.8	59
04	11	8	3	5	20–130	600	349.4	72.9	9.2	37
05	14	10	10	0	30–120	–	277.5	67.0	7.5	43
06	10	7	7	0	30–110	–	325.9	78.0	12.5	24
07	11	7	7	0	20–110	–	296.9	79.3	5.7	112
08	10	10	10	0	30–110	450–530	285.2	70.3	6.5	57
09	13	10	5	5	20–130	–	293.6	67.2	5.3	85
10[c]	14	–	–	–	–	–	–	–	–	–
11[c]	17	–	–	–	–	–	–	–	–	–
12	10	6	6	0	30–130	–	313.3	73.4	10.9	39
13	15	13	1	12	20– 50	–	307.8	74.5	8.0	28
14	11	7	3	4	35–110	–	322.3	67.3	10.1	37
15	14	13	9	4	30– 60	400–530	315.9	71.5	8.0	28
16[c]	13	–	–	–	–	–	–	–	–	–
17[b]	10	8	2	6	10– 40	500–560	289.2	69.4	9.6	34
18	12	10	4	6	30–110	–	267.1	79.1	7.1	48
19	10	5	0	5	20– 70	–	300.3	65.1	7.8	97
20	12	11	11	0	40–130	560–645	320.5	68.0	6.9	45
21	10	7	7	0	30–130	530–665	308.6	57.8	10.9	24
Mean	21	18					305.6	72.1	3.9	82

[a] *N* is the number of specimens measured per site; *n* is the number of specimens used in calculation of mean remanence directions; *e* is the number of specimens with directions obtained from endpoints; *r* is the number of specimens with directions obtained from remagnetization circles. AF demagnetization steps are ranges in milliTesla (mT); thermal demagnetization steps are ranges in degrees Celsius (°C). Mean remanence directions are given as declination (*D*), inclination (*I*), radius of 95% confidence (α_{95}), and precision parameter (*k*) (Fisher, 1953).

[b] Sites with mean remanence direction converted from the anti-parallel direction.

[c] Sites excluded from calculation of unit mean.

clase; F — LiF; Cl — Tugtupite; O — by difference; and for plagioclase were: Si — Amelia albite; Al — anorthite; Fe — fayalite; Ca — anorthite; Na — Amelia albite; K — orthoclase; and, O — by difference.

Representative amphibole analyses, site-averaged compositions, and P–T–d data are given in Tables 2 and 3 and Figs. 7 and 8. Amphibole structural formulae are calculated on the basis of 13 cations, a method that predicts site occupancies well and gives reasonable estimates of Fe^{3+} and Fe^{2+} (Cosca et al., 1991; Harris et al., 1997, 1999). The plagioclase structural formulae are calculated on the basis of 32 oxygens. The amphiboles exhibit a small compositional range (Table 2; Fig. 7) and are classified as magnesiohornblendes, potassian tschermakites and potassian ferrotschermakites.

Preliminary average pressures per site range from 464 to 743 MPa, and temperatures range from 689 to 814°C (Table 3). Final temperature-corrected pressures yield a small range from 359 to 486 MPa (average 441 MPa), corresponding to depths from 12.9 to 17.5 km (average 15.8 km) (Table 3; Fig. 8), assuming a geobarometric gradient of 0.036 km/MPa using a crustal density of 2.8 g/cm^3. Corrections for topographic relief were not necessary because all sites were collected from tors of similar elevation across the batholith. The standard deviation on the average Al^{tot} and the corresponding error on the pressure estimates are all less than

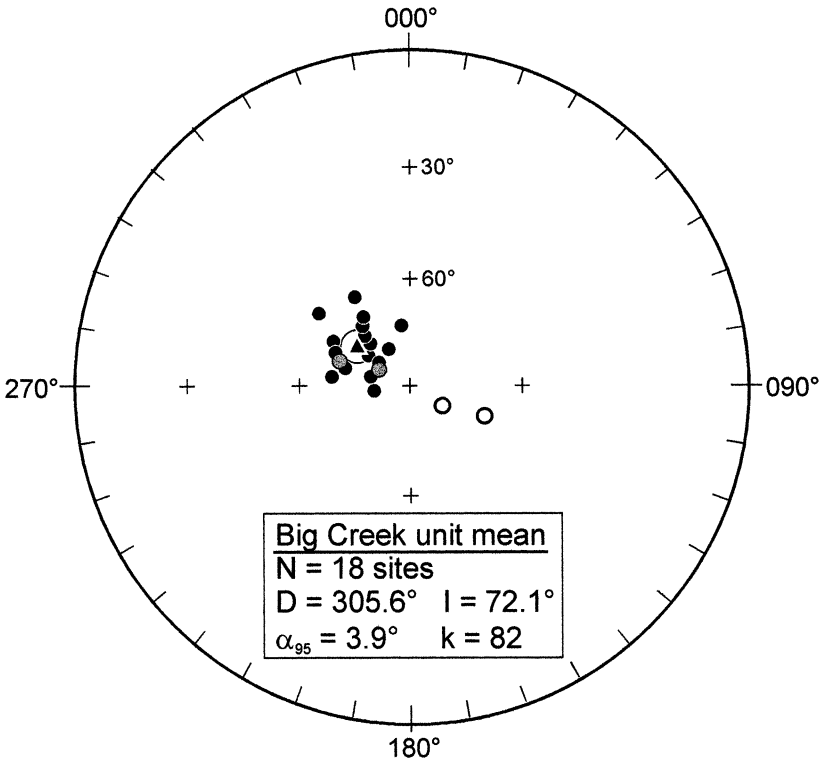


Fig. 6. Equal-area stereogram showing all site means of the Big Creek batholith (Table 1). Closed (open) symbols represent normal (reversed) polarity. The two gray symbols are the antipodal positions for the two reverse site ChRM directions. The unit mean is the triangle with its cone of 95% confidence.

the error in the calibration of the geobarometer (60 MPa; Anderson and Smith, 1995).

5. Discussion

Preliminary paleomagnetic results for the Late Cretaceous Seymour Creek stock have been reported by McCausland et al. (1998). This stock is an ovoid 4 km^2 intrusion of massive granodiorite to quartz monzonite that has given a U–Pb zircon date of 68.5 ± 0.3 Ma (J.K. Mortensen, pers. commun., 1999). Notably, the stock and associated dykes intrude the Big Creek batholith (Fig. 2). The stock retains a predominantly reversed ChRM in 16 sites (14 reversed- and two normal-polarity sites) that is directed steeply upwards, defining a paleopole position of 59°N, 197°E, ($d_p = 10°$, $d_m = 10°$) that is dissimilar from that of the Big Creek batholith. The remanence of the stock is a primary thermoremanent magnetization residing in magnetite because the pluton postdates regional deformation and is younger than other metamorphic and thermal events. Preliminary geobarometric results indicate that the stock was emplaced at a depth of 8.9 km. Therefore, a ChRM direction in magnetite with laboratory unblocking temperatures of >430°C would have been locked into the rock immediately on emplacement (Pullaiah et al., 1975; Dunlop, 1990), and virtually all sites retain their ChRM up to temperatures close to 575°C.

The pole positions for the $\sim$190 Ma Big Creek batholith and $\sim$69 Ma Seymour Creek stock are shown in Fig. 9a in relation to the apparent polar wander path (APWP) for the North American craton of Besse and Courtillot (1991). Also shown are their corresponding co-eval reference poles of

Table 2
Average amphibole analyses from the border zone of the Big Creek batholith[a]

Site	BC06	BC09	BC13	BC15	BC17	BC18	BC19
SiO_2	43.67	43.76	40.42	40.96	41.73	41.18	41.47
TiO_2	1.13	1.12	1.15	1.34	0.80	1.07	1.60
Al_2O_3	8.95	9.09	12.15	11.39	10.65	10.68	10.13
Cr_2O_3	0.00	0.03	0.00	0.00	0.02	0.00	0.00
FeO	17.34	16.74	20.22	19.76	18.82	19.39	18.70
MnO	0.59	0.57	0.73	0.79	0.45	0.55	0.54
MgO	10.95	11.21	7.89	8.32	9.60	9.23	9.23
CaO	11.78	11.99	11.87	11.83	11.83	11.66	11.65
Na_2O	1.45	1.41	1.15	1.24	1.47	1.57	1.58
K_2O	1.08	1.09	1.49	1.45	1.36	1.56	1.56
F	0.44	0.46	0.20	0.18	0.32	0.35	0.31
Cl	0.13	0.14	0.06	0.06	0.10	0.27	0.27
Total	97.52	97.61	97.34	97.32	97.15	97.50	97.05
Si	6.568	6.583	6.189	6.265	6.359	6.297	6.381
$^{[4]}Al$	1.432	1.417	1.811	1.735	1.641	1.703	1.619
$^{[6]}Al$	0.159	0.185	0.383	0.320	0.271	0.223	0.219
Fe^{3+}[b]	0.575	0.510	0.637	0.578	0.621	0.645	0.406
Fe^{2+}	1.604	1.582	1.953	1.950	1.777	1.835	2.001
Ti	0.131	0.122	0.132	0.154	0.092	0.123	0.185
Mg	2.455	2.525	1.800	1.896	2.179	2.103	2.118
Mn	0.075	0.072	0.095	0.103	0.058	0.072	0.070
Cr	0.000	0.004	0.000	0.000	0.002	0.000	0.000
Ca	1.898	1.930	1.947	1.938	1.932	1.910	1.921
$^{[M4]}Na$	0.102	0.070	0.053	0.062	0.068	0.090	0.079
$^{[A]}Na$	0.327	0.338	0.289	0.307	0.366	0.375	0.394
K	0.211	0.205	0.291	0.283	0.265	0.304	0.307
OH[b]	1.755	1.747	1.888	1.898	1.818	1.762	1.779
F	0.211	0.219	0.096	0.087	0.155	0.169	0.151
Cl	0.035	0.034	0.017	0.014	0.027	0.069	0.070
Fe^{3+}[c]	0.26	0.24	0.25	0.23	0.26	0.26	0.17
Fe[d]	0.47	0.45	0.59	0.57	0.52	0.54	0.53

[a] Structural formulae are normalized to 13 cations exclusive of Ca, Na and K.
[b] Back-calculated from the structural formula.
[c] $Fe^{3+}/(Fe^{2+}+Fe^{3+})$.
[d] $(Fe^{2+}+Fe^{3+})/(Fe^{2+}+Fe^{3+}+Mg)$.

71.0°N, 93.2°E ($A_{95}=3.9°$) at 190 Ma and of 72.3°N, 192.7°E ($A_{95}=4.1°$) at 69 Ma. When the indicated tectonic motions are calculated, the Big Creek batholith is $9.3\pm6.1°$ near-sided (NS) and rotated $32\pm11°$ counterclockwise (CCW), and the Seymour Creek stock is $2.0\pm8.6°$NS and rotated $57\pm 42°$CCW. Given that the stock retains a primary TRM and is concordant in paleolatitude for its present location (i.e. non-significantly near-sided, indicating translation neither towards nor away from the reference pole), a closer look at the Big Creek result is warranted. Its geobarometry indicates emplacement at a depth of ~16 km, which, assuming a geothermal gradient of ~33°C/km for this part of the YTT (T.J. Lewis, pers. commun., 1999), implies a host-rock temper-

Fig. 7. Hornblende compositions of the Big Creek batholith. (a) Calcic-amphibole classification plot (Leake et al., 1997). (b, c) Amphibole exchange reactions. Pressures shown in (c) are based on the Schmidt (1992) calibration.

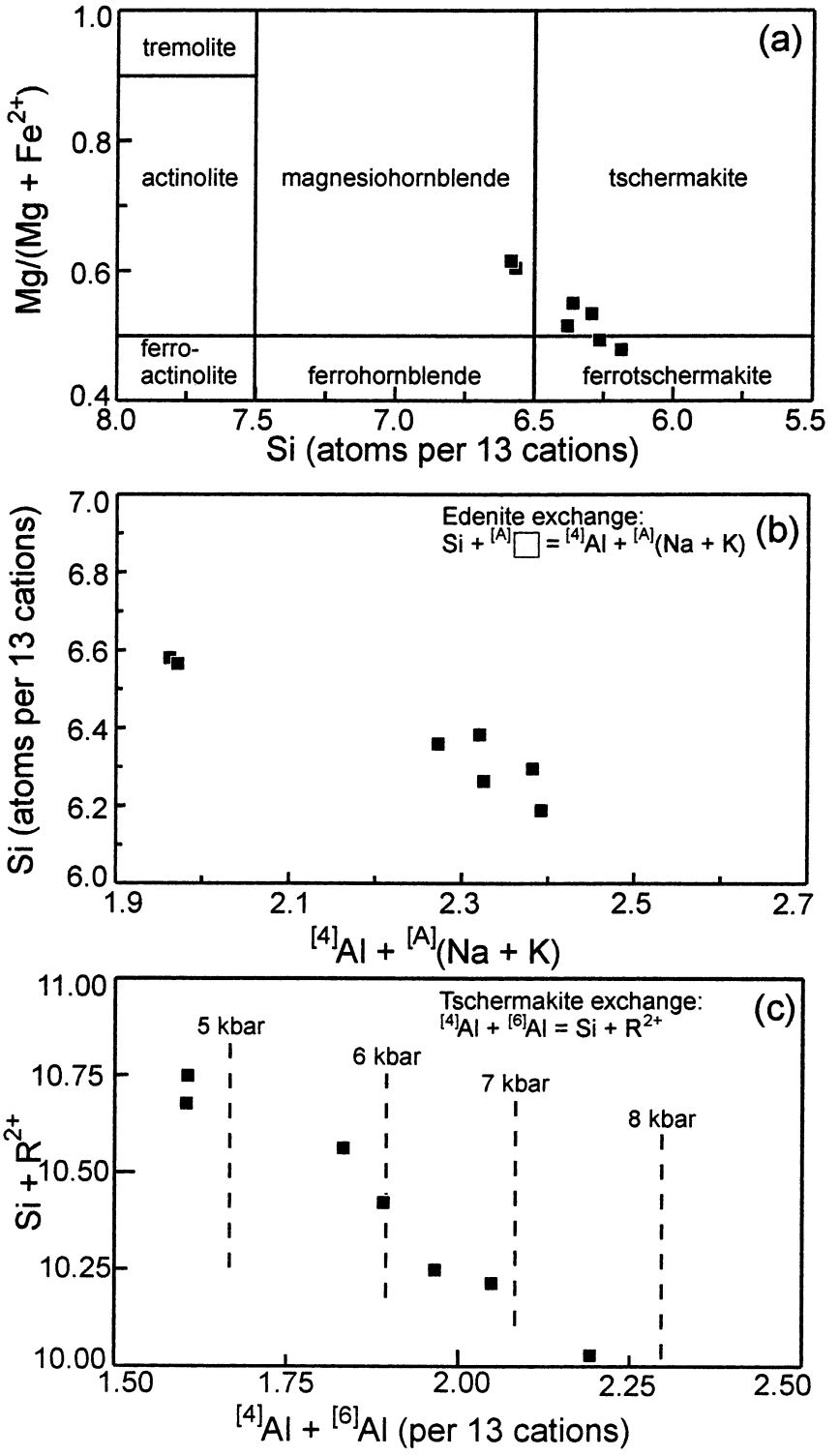

(a)
tremolite
actinolite
magnesiohornblende
tschermakite
ferro-
actinolite
ferrohornblende
ferrotschermakite
Mg/(Mg + Fe2+)
Si (atoms per 13 cations)
(b)
Edenite exchange:
Si + [A]□ = [4]Al + [A](Na + K)
Si (atoms per 13 cations)
[4]Al + [A](Na + K)
(c)
Tschermakite exchange:
[4]Al + [6]Al = Si + R2+
5 kbar
6 kbar
7 kbar
8 kbar
Si + R2+
[4]Al + [6]Al (per 13 cations)

Table 3
Site-averaged compositions of co-existing amphibole and plagioclase edges and geothermobarometric data for the Big Creek batholith

Site	Plag (Ab)	Amph (avg. Al^{tot})	P (Sch) (MPa)	T (B&H) (°C)	P (A&S) (MPa)	Depth (km)
BC06	72.7	1.591 ± 0.105	464	743	359	12.9
BC09	73.8	1.602 ± 0.060	465	689	447	16.1
BC13	61.5	2.194 ± 0.017	743	814	434	15.6
BC15	64.6	2.055 ± 0.020	677	797	428	15.4
BC17	74.6	1.912 ± 0.060	602	760	464	16.7
BC18	76.9	1.926 ± 0.073	636	761	466	16.8
BC19	76.7	1.838 ± 0.083	709	733	486	17.5

P(Sch) is from Schmidt (1992), T(B&H) is from Blundy and Holland (1990), and P(A&S) is from Anderson and Smith (1995). Plag Ab# is the atomic ratio [Na/(Na + Ca)] in plagioclase.

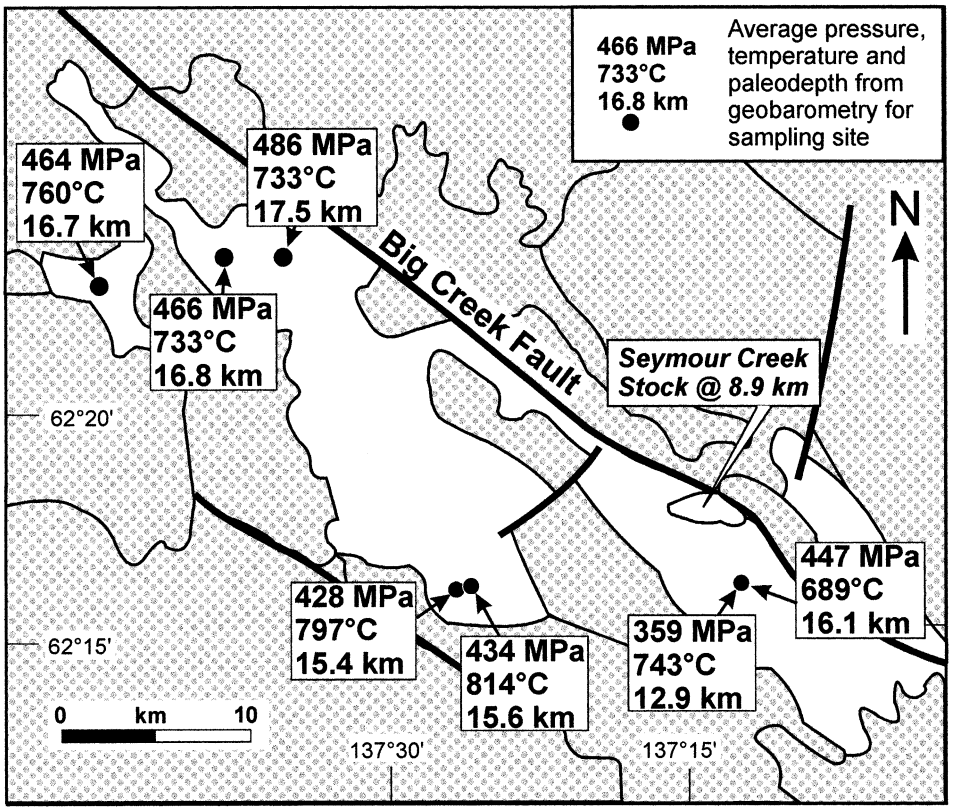


Fig. 8. Pressure, temperature and intrusion depth data from Table 3.

ature of ~530°C, approaching the 585°C Curie temperature of the magnetite in the batholith. Noting that the thermal demagnetization curves show that Big Creek's ChRM is retained through the 540–570°C range in many sites, application of the Pullaiah et al. (1975) curves indicates that the batholith had to be exhumed to a depth of ~14 km, again assuming a geothermal gradient

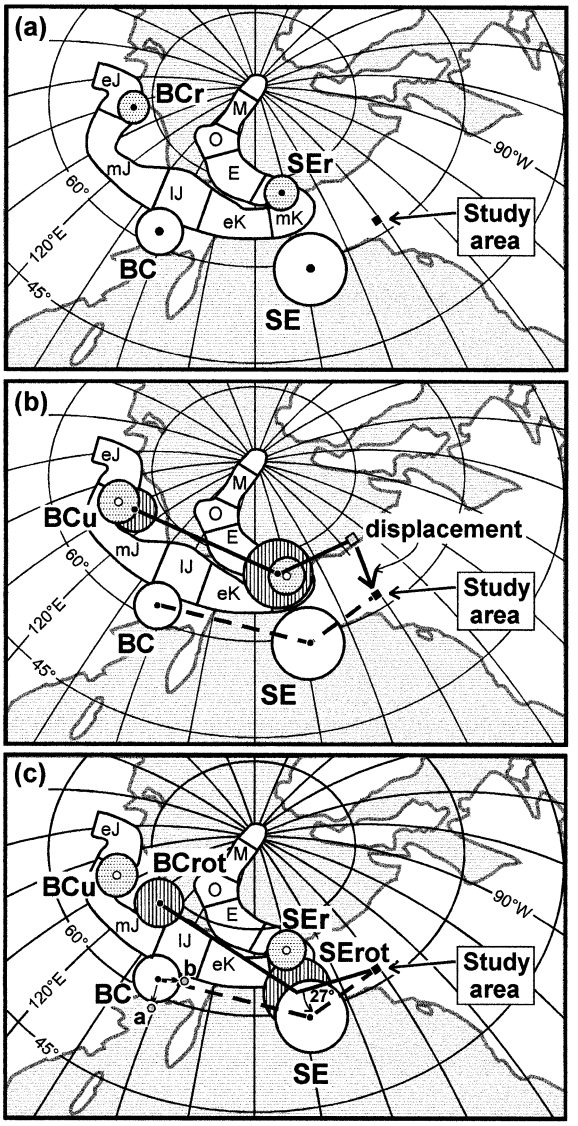


Fig. 9. (a) Locations of the Big Creek–Seymour Creek plutons' study area, their paleopoles and North American cratonic and reference poles, and the APWP for North America of Besse and Courtillot (1991) with their 1σ error limits. BCr: Big Creek 190 Ma reference pole; BC: Big Creek paleopole; SEr: Seymour Creek 70 Ma reference pole; SE: Seymour Creek paleopole. (b) Simple translation of the paleopoles onto their respective reference poles showing the possible location of the study area at 180 Ma. Abbreviations as in (a) except BCu: Big Creek uplift-age 180 Ma reference pole; (c) Simple rotation of the paleopoles using the 180 Ma rotation value. Abbreviations as in (a) and (b) except BCrot: Big Creek rotated paleopole; SErot: Seymour Creek rotated paleopole; a, b are corrected poles discussed in text.

of 33°/km, before its ChRM would be frozen into the rock as an uplift TRM.

Although the age of unburial is uncertain, three lines of evidence suggest that it was soon after emplacement of the Big Creek batholith. First, the K–Ar hornblende isotopic system is typically closed at ~530°C and this system has given ages of 188.8 ± 3.8 Ma and 184 ± 7 Ma for the batholith (Stevens et al., 1982; Hunt and Roddick, 1987). Thus, it is likely that the ChRM was set by 180 Ma. Second, U–Pb zircon and titanite isotopic determinations for the nearby and coeval Aishihik and Long Lake plutonic suites in the YTT indicate rapid uplift from ~189 to ~186 Ma with cooling through ~500°C by 180 Ma (Johnston and Erdmer, 1995; Johnston et al., 1996a). Third, it is reasonable to expect rapid orogenic uplift in the YTT given the extensive emplacement of large plutons during the Early Jurassic and the Jurassic regional cooling ages. Therefore, using the 180 Ma reference pole from Besse and Courtillot (1991) of 101.2°E, 68.1°N ($A_{95} = 2.7°$) results in tectonic motions for the Big Creek batholith terrane of $10.1 \pm 5.6°$ NS and $27 \pm 10°$ CCW.

The ~190 Ma Big Creek batholith is partially overlain, and at one time may have been totally overlain, by the flat-lying to gently dipping 70 ± 2 Ma Carmacks Group of subaerial volcanic rocks. Wynne et al. (1998) have provided paleomagnetic data for the Carmacks Group, upgrading an earlier study by Marquis and Globerman (1988), and for the underlying ~185 Ma Early Jurassic Nordenskiöld Formation tuffs and ~210 Ma Late Triassic Mandanna Member redbeds. They found that all three units retain a bipolar ChRM with mean directions of $D = 346.3°$, $I = 69.3°$ ($\alpha_{95} = 4.9°$, $N = 26$, tilt corrected, positive fold test), $D = 323.4°$, $I = 82.6°$ ($\alpha_{95} = 6.6°$, $N = 15$, uncorrected), and $D = 338.8°$, $I = 69.5°$ ($\alpha_{95} = 4.8°$, $N = 14$, uncorrected, negative fold and conglomerate tests, positive Eocene contact test). Wynne et al. (1998) interpreted their results to show that the Nordenskiöld and Mandanna units were deformed during the Middle to Late Jurassic and then remagnetized by subvolcanic hydrothermal fluid flows when the Carmacks Group volcanic rocks were extruded on top of them at 70 ± 2 Ma. They go on to suggest that older rocks in the southern Yukon and northern British Columbia

were subject to a regional remagnetization coeval with Carmacks magmatism. We reject the hypothesis on: (1) a local basis because the Big Creek and Seymour Creek ChRM directions are significantly different at $\gg 95\%$ confidence from the Carmacks and Mandanna directions, as is the Nordenskiöld direction, and (2) a regional basis because virtually all older units measured to date in the region give statistically significantly different directions (Harris et al., 1999). The explanation may be that red beds and tuffs are inherently porous and permeable rock types that are very amenable to hydrothermal alteration, whereas the Big Creek and Seymour Creek plutons, and most other rock units providing paleomagnetic data in the region, are composed of massive igneous rocks that are not appreciably amenable to hydrothermal alteration.

The discordance between the measured Big Creek and Seymour Creek poles in the YTT and the cratonic reference poles can be accounted for in two reasonable ways. The preferred way is to consider that they define a segment of the APWP from 180 to 70 Ma. When this segment is then aligned in its 'best-fit' position on the APWP, the study area moves correspondingly from its present position of 222.5°E, 62.3°N to an Early Jurassic position relative to the craton (Fig. 9b). Relative to the craton, therefore, this calculation indicates that the YTT was translated southward by $\sim 12^\circ$ and rotated $\sim 13^\circ$ CCW. It should be noted, however, that a simple rotation of $\sim 27^\circ$ CCW also gives a statistically reasonable fit for the YTT in the study area (Fig. 9c). Thus, these data provide initial an indication that the Big Creek area of the YTT cannot be a far-travelled allochthonous terrane since the Early Jurassic. Rather, they show speculatively that it is a parautochthon, being only slightly displaced and/or slightly rotated CCW about a near-vertical axis.

Perhaps the most surprising feature of this initial paleomagnetic study of plutons in the YTT is that the results are diametrically different from previous studies on Cretaceous rocks in the Intermontane and Coast belts of Canada and southeastern Alaska. For example, Irving et al. (1996) have summarized the Cretaceous data. They showed that the four Cretaceous poles from rock units in the Intermontane Belt are far-sided by an average of $\sim 13 \pm 4^\circ$ or $\sim 1500 \pm 500$ km and rotated by $\sim 50 \pm 29^\circ$ clockwise (CW) and that the 13 poles from the Coast Belt are far-sided by an average of $\sim 27 \pm 8^\circ$ and rotated by $\sim 39^\circ \pm 20^\circ$ CW. Several recent poles from Cretaceous and Jurassic rocks in the Intermontane terranes of the Yukon also have given far-sided and clockwise-rotated poles (Harris et al., 1999; Symons et al., 2000, this issue). The obvious interpretation of these different paleomagnetic data sets is that the YTT is not far-travelled and acted as the 'bumper' and ramp for the North American craton against which, and over which, the accreted terranes of the Intermontane and Coast belts travelled until they were accreted. One test of this model is that it predicts that large-scale sole faults will be imaged between the YTT and overlying allochthonous Intermontane terranes by LITHOPROBE's deep-crustal seismology along SNORCLE lines 2 and 3 (Clowes, 1997). Ultimately, of course, surface mapping and drilling should find these faults, and geological evidence such as cross-bed and sole mark orientations should be found to reflect the indicated rotations on the sole faults (Symons et al., in press).

There are other lines of evidence, albeit somewhat nebulous, that suggest that the YTT has been proximal to its current location since Early Jurassic. One, obviously, is the noted affinity of YTT rock types to North American units (Tempelman-Kluit, 1984; Mortensen, 1992). A second is the significant differences in magnetic seismic and gravity data at the Tintina fault boundary (Lowe et al., 1994). Another is the mismatch in age and Sr isotopic signatures between the clasts in the Laberge Group Conglomerate Formation of the northern Stikine Terrane and the plutons to the immediate west in the YTT that should have been their source if the two terranes were abutting and travelling together (Hart et al., 1995), suggesting that the two terranes were separated when the conglomerate was deposited. Geochemical evidence from Cretaceous sub-alkaline granitoid plutons across the Yukon indicates primitive Sr and Nd isotope values across the Stikine Terrane and YTT indicating a primary arc origin. In contrast, more evolved values are found for granitoids in the Cassiar Terrane and eastward that indicate the likelihood of a Precambrian cratonic basement below (Morris and Creaser, 1999). Similarly, Francis et al. (1999) report a fundamental change

in the U, Sr and Nd isotopes and in various elemental ratios of Late Cretaceous to Recent volcanics close to the eastern limit of the YTT. From the analysis of geothermal data, Hyndman and Lewis (1999) also predict that the Precambrian cratonic margin extends westward to about the YTT eastern boundary, requiring a very different or continental crust such as the YTT to be under the Intermontane terranes.

At this preliminary stage in YTT paleomagnetism, two caveats need stating. One is that the results to date represent only one small area of the YTT. They could prove anomalous. However, several other studies are in progress that will substantially expand the paleomagnetic coverage both areally and in geologic time. Until a consistent pattern for the YTT pole positions is established, the pericratonic geotectonic model should be regarded as an hypothesis. The second caveat is that obvious factors have yet to be built into the estimate of terrane motion. For example, Gabrielse (1985) has provided an estimate of as much as $\sim$800 km of dextral displacement on the Tintina Fault (Fig. 1) in the Yukon that postdates Middle Jurassic time. Correcting the Big Creek pole for this motion moves it by $\sim 7^\circ$ southerly (Fig. 9c, pole a) with corresponding minor changes in the rotation and displacement estimates. Similarly, although the geobarometric estimates are not statistically different, they do imply an $\sim$2 km difference along the $\sim$40 km northwest–southeast sampled length of the batholith. It in turn implies $\sim 3^\circ$ SE-down tilt on exhumation of the batholith that, if corrected for, would move the pole $\sim 8^\circ$ easterly (Fig. 9c, pole b), implying a younger age of remanence acquisition of $\sim$145 Ma with concomitant minor changes in the tectonic motion estimates. Nevertheless, it is still certain that the YTT results given by the Big Creek batholith are radically different from those of the terranes of the Coast and Intermontane belts, and they imply a very different geotectonic history.

Acknowledgements

The authors thank M.T. Smethurst for helping with the sampling of the collection, Yves Thibault for his help with the microprobe analyses, the Smithsonian Institute of Science for the geothermobarometric standards (USNM designated), Mike Jackson and Randy Enkin for many useful suggestions as referees, and the Natural Sciences and Engineering Research Council for funding through LITHOPROBE grants. This is LITHOPROBE publication number 1089.

References

Anderson, J.L., Smith, D.R., 1995. The effects of temperature and f_{O2} on the Al-in-hornblende barometer. Am. Miner. 80, 549–559.

Besse, J., Courtillot, V., 1991. Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian Plates, and true polar wander since 200 Ma. J. Geophys. Res. 96, 4029–4050.

Blundy, J.D., Holland, T.J.B., 1990. Calcic amphibole equilibria and a new amphibole-plagioclase geothermometer. Contrib. Miner. Petrol. 104, 208–224.

Carlson, G.G., 1987. Geology of Mount Nansen (115I/3) and Stoddart Creek (115I/6) map areas, Dawson Range, central Yukon. Exploration and Geological Services Division. Indian and Northern Affairs Canada, Yukon, Open File 1987-2. 181 pp.

Clowes, R.M. (Ed.), 1997. LITHOPROBE Phase V Proposal — Evolution of a Continent Revealed. Lithoprobe Secretariat, The University of British Columbia, Vancouver, BC, 292 pp

Cosca, M.A., Essene, E.J., Bowman, J.R., 1991. Complete chemical analyses of metamorphic hornblendes: implications for normalizations, calculated H_2O activities and thermobarometry. Contrib. Miner. Petrol. 108, 472–484.

Dunlop, D.J., 1990. Developments in rock magnetism. Rep. Prog. Phys. 53, 707–792.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. Roy. Soc. Lond. A 217, 295–305.

Francis, D., Abraham, A.C., Polve, M., 1999. Volcanic probes of the craton's edge beneath the northern Canadian Cordillera. In: LITHOPROBE Report No. 69, 142–146.

Gabrielse, H., 1985. Major dextral transcurrent displacements along the northern Rocky Mountain Trench and related lineaments in north-central British Columbia. Geol. Soc. Am. Bull. 96, 1–14.

Halls, H.C., 1978. The use of converging remagnetization circles in paleomagnetism. Phys. Earth Planet. Inter. 16, 1–11.

Hansen, V.L., 1990. Yukon–Tanana terrane: A partial acquittal. Geology 18, 365–369.

Hansen, V.L., Hietzler, T.M., Harrison, T.M., 1991. Mesozoic thermal evolution of the Yukon–Tanana terrane: New evidence from ^{40}Ar–^{39}Ar data. Tectonics 10, 51–76.

Hansen, V.L., Dusel-Bacon, C., 1998. Structural and kinematic evolution of the Yukon–Tanana upland tectonites, east-central Alaska: A record of late Paleozoic to Mesozoic crustal assembly. GSA Bull. 110, 211–230.

Harris, M.J., Symons, D.T.A., Blackburn, W.H., Hart, C.J.R., 1997. Paleomagnetic and geobarometric study of the mid Cretaceous Whitehorse pluton, Yukon Territory. Can. J. Earth Sci. 34, 1379–1391.

Harris, M.J., 1998. Geotectonic motions of northern Canadian Cordillera terranes from paleomagnetic measurements with geobarometric corrections. Ph.D. thesis, University of Western Ontario, London, Ontario.

Harris, M.J., Symons, D.T.A., Blackburn, W.H., Hart, C.J.R., 1999. Paleomagnetic and geobarometric study of the Late Cretaceous Mount Lorne Stock, southern Yukon Territory. Can. J. Earth Sci. 36, 905–915.

Hart, C.J.R., Dickie, J.R., Ghosh, D.K., Armstrong, R.L., 1995. Provenance constraints for Whitehorse Trough conglomerate: U–Pb zircon dates and initial Sr ratios of granitic clasts in Jurassic Laberge Group, Yukon Territory. In: Miller, D.M., Busby, C. (Eds.), Jurassic Magmatism and Tectonics of the North American Cordillera. Geol. Soc. Am., Spec. Pap. 299, 47–63.

Hunt, P.A., Roddick, J.C., 1987. A compilation of K–Ar ages.In: Report 17, Geological Survey of Canada, 143–210.

Hyndman, R.D., Lewis, T.J., 1999. High heat flow along the SNORCLE transect. In: LITHOPROBE Report 69, 83–84..

Irving, E., Wynne, P.J., Thorkelson, D.J., Schiarizza, P., 1996. Large (1000 to 4000 km) northward movements of tectonic domains in the northern Cordillera, 83 to 45 Ma. J. Geophys. Res. 101, 17901–17916.

Johnston, S.T., Erdmer, P., 1995. Hot-side-up aureole in southwest Yukon and limits on terrane assembly of the northern Canadian Cordillera. Geology 23, 419–422.

Johnston, S.T., Mortensen, J.K., Erdmer, P., 1996a. Igneous and metaigneous age constraints for the Aishihik metamorphic suite, southwest Yukon. Can. J. Earth Sci. 33, 1543–1555.

Johnston, S.T., Wynne, P.J., Francis, D., Hart, C.J.R., Enkin, R.J., Engebretson, D.C., 1996b. Yellowstone in Yukon: The Late Cretaceous Carmacks Group. Geology 24, 997–1000.

Kirschvink, J.L., 1980. The least squares line and plane and the analysis of paleomagnetic data. Geophys. J. Roy. Astron. Soc. 62, 699–718.

Leake, B.E., et al., 1997. Nomenclature of amphiboles: Report of the Subcommittee on Amphiboles of the International Mineralogical Association, Commission on New Minerals and Mineral Names. Can. Miner. 35, 219–246.

Lowe, C., Horner, R.B., Mortensen, J.K., Johnston, S.T., Roots, C.F., 1994. New geophysical data from the northern Cordillera: preliminary interpretations and implications for the tectonics and deep geology. Can. J. Earth Sci. 31, 891–904.

Marquis, G., Globerman, B.R., 1988. Northward motion of the Whitehorse Trough: Paleomagnetic evidence from the Upper Cretaceous Carmacks Group. J. Earth Sci. 25, 2005–2016.

McCausland, P.J.A., Symons, D.T.A., Hart, C.J.R., 1998. Post-Cretaceous geotectonic motion of the Yukon–Tanana terrane from paleomagnetism of the Seymour Creek stock, Yukon Territory. In: Abstract, Eos Trans. AGU, Fall Mtg. Suppl. 75, F219–F220.

McClelland, W.C., Gehrels, G.E., Saleeby, J.B., 1992. Upper Jurassic–Lower Cretaceous basinal strata along the Cordilleran margin: Implications for the accretionary history of the Alexander–Wrangellia–Peninsular terrane. Tectonics 11, 823–835.

McInnes, B.I.A., 1987. Geological and precious metal evolution at Freegold Mountain, Dawson Range, Yukon. M.Sc. thesis, McMaster University, Hamilton.

Mihalynuk, M.G., Nelson, J.L., Diakow, L.J., 1987. Cache Creek terrane entrapment: Oroclinal paradox within the Canadian Cordillera. Tectonics 13, 575–595.

Morris, G.A., Creaser, R.A., 1999. Mid-Cretaceous granites of the southern Yukon: Initial report. In: LITHOPROBE Report 69, 120–121.

Mortensen, J.K., Jilson, G.A., 1985. Evolution of the Yukon–-Tanana Terrane: Evidence from southeastern Yukon Territory. Geology 13, 806–810.

Mortensen, J.K., 1992. Pre-mid-Mesozoic evolution of the Yukon–Tanana Terrane, Yukon and Alaska. Tectonics 11, 836–853.

Pavlis, T.L., Sisson, V.B., Foster, H.L., Nokleberg, W.J., Plafker, G., 1993. Mid-Cretaceous extensional tectonics of the Yukon–Tanana Terrane, Trans-Alaska Crustal Transect (TACT), east-central Alaska. Tectonics 12, 103–122.

Payne, J.G., Gonzalez, R.A., Akhurst, K., Sisson, W.G., 1987. Geology of Colorado Creek (115-J/10), Selwyn River (115-J/9), and Prospector Mountain (115-I/5) map areas, western Dawson Range, west-central Yukon. In: Expl. Geol. Serv. Div., Yukon, Indian and Northern Affairs Canada, 35–38.

Pullaiah, G.E., Irving, E., Buchan, K.L., Dunlop, D.J., 1975. Magnetization changes caused by burial and uplift. Earth Planet. Sci. Lett. 28, 133–143.

Schmidt, M.W., 1992. Amphibole composition in tonalite as a function of pressure: an experimental calibration of the Al-in-hornblende barometer. Contrib. Miner. Petrol. 110, 304–310.

Stevens, R.D., DeLabio, R.N., LaChance, G.R., 1982. Age determinations and geological studies, K–Ar isotopic ages. In: Report 15, Geological Survey of Canada, 17–27.

Symons, D.T.A., Harris, M.J., Gabites, J.E., Hart, C.J.R., 2000. Eocene (51 Ma) end to northward translation of the Coast Plutonic Complex: Paleomagnetism and K–Ar dating of the White Pass dikes. Tectonophysics 326, 93–108.

Templeman-Kluit, D.J., 1976. The Yukon Crystalline Terrane, enigma in the Canadian Cordillera. Geol. Soc. Am. Bull. 87, 1343–1357.

Tempelman-Kluit, D.J., 1984. Geology, Laberge (105E) and Carmacks (115I), Yukon Territory. Geological Survey of Canada, Open File 1101, Scale 1:250 000.

Tempelman-Kluit, D.J., Gabrielse, H., Evenchick, C.A., et al., 1991. Part D. Omineca Belt. In: Gabrielse, H., Yorath, C.J. (Eds.), Geology of the Cordilleran Orogen in Canada. Geological Survey of Canada, Geology of Canada, No. 4, 594–634., Chapter 17.

Wynne, P.J., Enkin, R.J., Baker, J., Johnston, S.T., Hart, C.J.R., 1998. The big flush: paleomagnetic signature of a 70 Ma regional hydrothermal event in displaced rocks of the northern Canadian Cordillera. Can. J. Earth Sci. 35, 657–671.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.